

Generating Indistinguishable Single-Photons from Strongly Dissipative Quantum Emitters using Cavity-Funneling in GaN

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1 Introduction

2 Theory

all the first part will be about this paper (cite)

Follow Thomas Grange's work [1]. Pure dephasing makes solid states single photons emitters difficult outside of cryogenic temperatures. we can filter the emission to counter that but low efficiency. Current Purcell effects aren't enough. Note that dephasing makes difficult because it widens the emission spectra So we try emitter cavity coupling in the right regime for enhanced emission in the right mode. We want to find the right regime, so first is solving and plotting. First we make following approximations (explain markovian etc) We are interested in I of course, also in beta because we want to be able to exploit. Start with beta plot because easier, as training with qutip.

explain gamma, kappa, g

add explanation of basis (why 3 dimension etc)

2.1 The Lindblad master equation

LME is useful for open systems

We initialise the system with one excitation in the emitter, $\rho_0 = |e, 0\rangle\langle e, 0|$, and evolve it under the Lindblad master equation:

$$\frac{d\rho}{dt} = \mathcal{L}[\rho] = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right) \quad (1)$$

with $H = \hbar g(|e, 0\rangle\langle g, 1| + |g, 1\rangle\langle e, 0|)$ the system Hamiltonian describing the unitary aspects of the dynamics (in this case the emitter-cavity coupling), and L_k the three collapse operators encoding the dissipative channels:

- $L_1 = \sqrt{\gamma} |g, 0\rangle\langle e, 0|$ spontaneous emission (bad channel)
- $L_2 = \sqrt{\kappa} |g, 0\rangle\langle g, 1|$ cavity decay into (good channel)
- $L_3 = \sqrt{\gamma^*} |e, 0\rangle\langle e, 0|$ pure dephasing (no population change)

3 Efficiency

The efficiency β is defined as the probability that a photon, following the initial excitation of the emitter, is emitted through the cavity's good channel rather than lost to the environment. Formally, it is the total flux of photons leaving through the cavity mode:

$$\beta = \kappa \int_0^\infty \langle \hat{n}_{\text{cav}}(t) \rangle dt = \kappa \int_0^\infty \text{Tr}[P_{\text{cav}} \rho(t)] dt \quad (2)$$

where $P_{\text{cav}} = |g, 1\rangle\langle g, 1|$ is the projector onto the cavity-occupied state and κ is the cavity decay rate. Physically, this makes a lot of sense as it is intuitively the average number of photons in the cavity over time, weighted by κ , the rate at which each photon escapes through the good channel.

3.1 Numerical approach

describe the QuTiP simulation, convergence study, and plots
attempted solutions, mesh etc

Failure of the numerical approach

The numerical simulation using QuTiP's master-equation solver (`mesolve`) failed to converge for a significant portion of the (g, κ) parameter grid. The root cause is *stiffness*: the system involves timescales that differ by many orders of magnitude. Specifically, the pure dephasing rate $\gamma^* = 10^4 \gamma$ is much larger than γ , while at the same time large values of g or κ introduce respectively rapid Rabi oscillations or fast cavity decay. Resolving the fastest timescale while integrating over the slowest one demands an extremely large number of time steps, making the simulation computationally intractable for the upper-right corner of the parameter space. This observation was the motivation to start looking for an analytical solution to the problem.

check the timescale discussion

explain the attempts at having lower tolerance for both, fix nt,

3.2 Analytical approach

3.2.1 Vectorisation of the density matrix

explain the logic of vectorisation and motivation (linear) + cite the ref in code

The density matrix can be flattened into a column vector by stacking its columns:

$$\rho \mapsto |\rho\rangle\rangle \in \mathbb{C}^{d^2} \quad (3)$$

3x3 to 1x9 - expliquer dimensions?

This turns the Lindblad equation into an ordinary linear ODE:

$$\frac{d}{dt} |\rho\rangle\rangle = \hat{\mathcal{L}} |\rho\rangle\rangle \implies |\rho(t)\rangle\rangle = e^{\hat{\mathcal{L}}t} |\rho_0\rangle\rangle \quad (4)$$

$\hat{\mathcal{L}}$ is a 9×9 complex matrix called the *Liouvillian superoperator*. Using the identity $|AXB\rangle\rangle = (B^\top \otimes A)|X\rangle\rangle$, it takes the explicit form:

$$\hat{\mathcal{L}} = -\frac{i}{\hbar}(I \otimes H - H^\top \otimes I) + \sum_k \left[L_k^* \otimes L_k - \frac{1}{2} \left(I \otimes L_k^\dagger L_k + (L_k^\dagger L_k)^\top \otimes I \right) \right] \quad (5)$$

Expectation values then become inner products in the vectorised space:

$$\text{Tr}[P_{\text{cav}} \rho(t)] = \langle\langle P_{\text{cav}} | \rho(t) \rangle\rangle \quad (6)$$

so that the efficiency (2) reads:

$$\beta = \kappa \int_0^\infty \langle\langle P_{\text{cav}} | e^{\hat{\mathcal{L}}t} | \rho_0 \rangle\rangle dt \quad (7)$$

3.2.2 The singular Liouvillian

All eigenvalues of $\hat{\mathcal{L}}$ have non-positive real parts, and there is a unique zero eigenvalue corresponding to the steady state $\rho_{\text{ss}} = |g, 0\rangle\langle g, 0|$ (all excitation has decayed). This makes $\hat{\mathcal{L}}$ singular, so one cannot directly invert the integral in (7).

explain standard fix: subtract steady state

$$\sigma(t) = \rho(t) - \rho_{\text{ss}}$$

decays to 0 + see hand notes for the other steps

$\frac{d}{dt} |\sigma\rangle\rangle = \hat{\mathcal{L}} |\sigma\rangle\rangle$. Integrating both sides from 0 to ∞ :

$$\hat{\mathcal{L}} \int_0^\infty |\sigma(t)\rangle\rangle dt = \left[e^{\hat{\mathcal{L}}t} |\sigma_0\rangle\rangle \right]_0^\infty = -|\sigma_0\rangle\rangle \quad (8)$$

where $|\sigma_0\rangle\rangle = |\rho_0 - \rho_{ss}\rangle\rangle$.

This is now invertible on the subspace orthogonal to the null space of L ?

$$\int_0^\infty |\sigma(t)\rangle\rangle dt = -\hat{\mathcal{L}}^{-1} |\sigma_0\rangle\rangle \quad (9)$$

Since $\text{Tr}[P_{\text{cav}} \rho_{ss}] = 0$ (there are no photons in the cavity at steady state), we have

$$\int_0^\infty \langle\langle P_{\text{cav}} | \rho(t) \rangle\rangle dt = \int_0^\infty \langle\langle P_{\text{cav}} | \sigma(t) \rangle\rangle dt \quad (10)$$

and the efficiency reduces to the closed-form expression:

$$\boxed{\beta = -\kappa \langle\langle P_{\text{cav}} | \hat{\mathcal{L}}^{-1} |\rho_0 - \rho_{ss}\rangle\rangle} \quad (11)$$

In practice this reduces to a single 9×9 linear solve $\hat{\mathcal{L}} x = -|\rho_0 - \rho_{ss}\rangle\rangle$, which is exact and can be computed numerically very fast, in a time independent of the values of g , κ , or γ^* .

3.3 Results

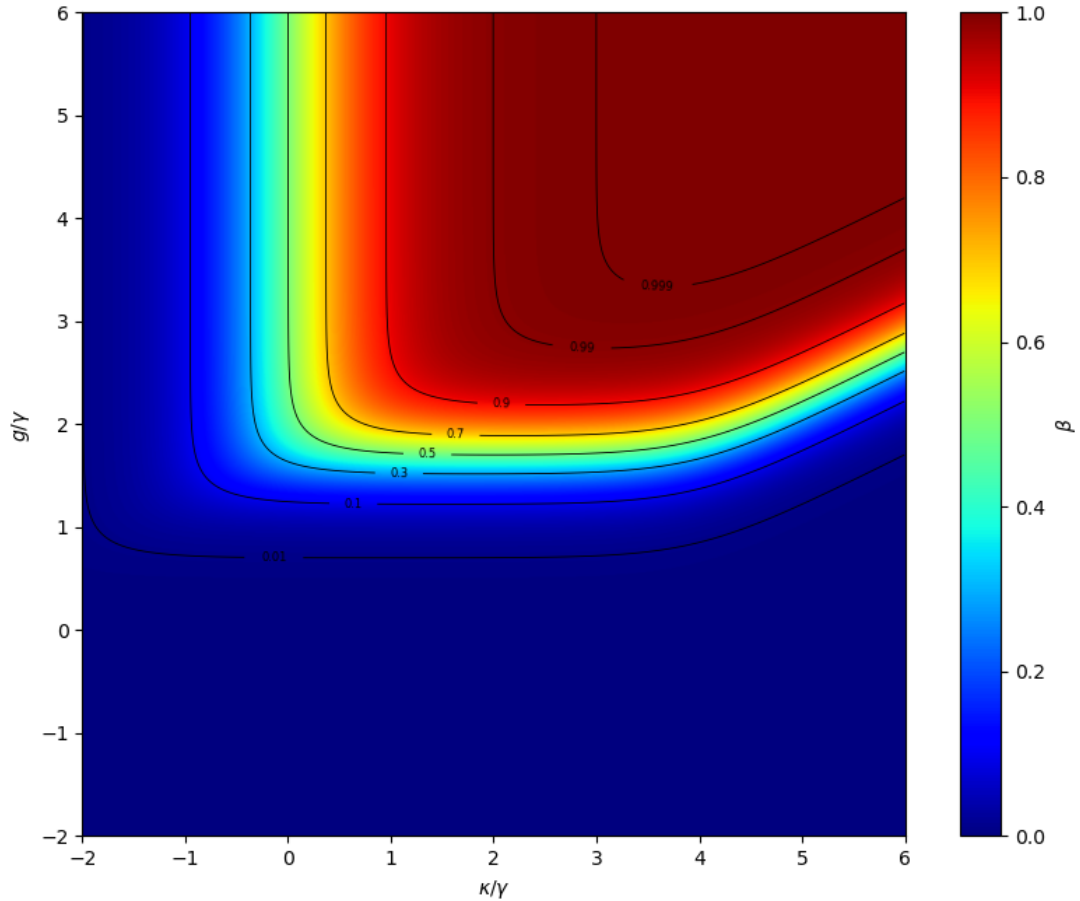


Figure 1: Efficiency β as a function of the coupling g and cavity decay rate κ , for a fixed ratio $\gamma^*/\gamma = 10^4$.

3.4 Remarks

In hindsight, it was possible to solve this more easily, without the vectorisation. However, the formalism is useful for the part on indistinguishability so we accept to pay that price, and it is

already done anyways.

Still, we can sketch the approach, as a system of ODE.

In the basis $\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle\}$, ρ is:

$$\rho = \begin{pmatrix} \rho_{gg} & \rho_{gc} & \rho_{ge} \\ \rho_{cg} & \rho_{cc} & \rho_{ce} \\ \rho_{eg} & \rho_{ec} & \rho_{ee} \end{pmatrix} \quad (12)$$

Each element evolves as $\dot{\rho}_{ij} = \langle i | \dot{\rho} | j \rangle$, given by the Lindblad equation (1).

First term: The commutator $[H, \rho]$ gives for each element:

$$\langle i | [H, \rho] | j \rangle = \langle i | H \rho | j \rangle - \langle i | \rho H | j \rangle \quad (13)$$

with, as a reminder, the Hamiltonian $H = \hbar g(|e, 0\rangle\langle g, 1| + |g, 1\rangle\langle e, 0|)$.

For example for $\dot{\rho}_{ee}$:

$$\begin{aligned} -\frac{i}{\hbar} \langle e, 0 | [H, \rho] | e, 0 \rangle &= -\frac{i}{\hbar} \left(\langle e, 0 | H \rho | e, 0 \rangle - \langle e, 0 | \rho H | e, 0 \rangle \right) \\ &= -\frac{i}{\hbar} \left(\sum_k \langle e, 0 | H | k \rangle \langle k | \rho | e, 0 \rangle - \sum_l \langle e, 0 | \rho | l \rangle \langle l | H | e, 0 \rangle \right) \\ &= -ig(\rho_{ce} - \rho_{ec}) \end{aligned}$$

Most terms vanish before the last step.

After some tedious computation that can be found in Appendix, we finally get

$$\begin{aligned} -\frac{i}{\hbar} [H, \rho]_{ee} &= -ig(\rho_{ce} - \rho_{ec}) \\ -\frac{i}{\hbar} [H, \rho]_{cc} &= +ig(\rho_{ce} - \rho_{ec}) \\ -\frac{i}{\hbar} [H, \rho]_{ec} &= +ig(\rho_{ee} - \rho_{cc}) \end{aligned}$$

ajouter le reste en appendix sauf si trop chiant, et mettre la ref

Second term: The contribution of each operator L_k is:

$$L_k \rho L_k^\dagger - \frac{1}{2} (L_k^\dagger L_k \rho + \rho L_k^\dagger L_k)$$

Projected onto element (i, j) :

$$\langle i | L_k \rho L_k^\dagger | j \rangle - \frac{1}{2} (\langle i | L_k^\dagger L_k \rho | j \rangle + \langle i | \rho L_k^\dagger L_k | j \rangle)$$

Take $L_1 = \sqrt{\gamma} |g, 0\rangle\langle e, 0|$ as an example.

First, $L_1^\dagger L_1 = \gamma |e, 0\rangle\langle e, 0| = \gamma P_e$, so the anticommutator term gives:

$$-\frac{\gamma}{2} \langle i | \{P_e, \rho\} | j \rangle = -\frac{\gamma}{2} (\langle i | P_e \rho | j \rangle + \langle i | \rho P_e | j \rangle)$$

And the $L_1 \rho L_1^\dagger$ term gives:

$$\gamma \langle i | g, 0 | i | g, 0 \rangle \langle e, 0 | \rho | e, 0 \rangle \langle g, 0 | j | g, 0 | j \rangle = \gamma \delta_{i,g0} \delta_{j,g0} \rho_{ee}$$

So L_1 drains ρ_{ee} into ρ_{gg} at rate γ , and kills all coherences involving $|e, 0\rangle$ at rate $\gamma/2$.

The same procedure is applied to $L_2 = \sqrt{\kappa} |g, 0\rangle\langle g, 1|$ and $L_3 = \sqrt{\gamma^*} |e, 0\rangle\langle e, 0|$.

si ça a un interet mettre dans annexe aussi le reste mais bon

Now, applying the dissipator contributions to each equations yields:

$$\begin{aligned}\dot{\rho}_{ee} &= -\gamma \rho_{ee} - ig(\rho_{ce} - \rho_{ec}) \\ \dot{\rho}_{cc} &= -\kappa \rho_{cc} + ig(\rho_{ce} - \rho_{ec}) \\ \dot{\rho}_{ec} &= -\frac{\gamma + \kappa + \gamma^*}{2} \rho_{ec} + ig(\rho_{ee} - \rho_{cc})\end{aligned}$$

By hermiticity, $\rho_{ce} = \rho_{ec}^*$, so $\rho_{ce} - \rho_{ec} = -2i \text{Im}(\rho_{ec})$. Defining $v = \text{Im}(\rho_{ec})$, the system simplifies to three real equations:

$$\begin{cases} \dot{\rho}_{ee} = -\gamma \rho_{ee} - 2g v \\ \dot{\rho}_{cc} = -\kappa \rho_{cc} + 2g v \\ \dot{v} = g(\rho_{ee} - \rho_{cc}) - \frac{\gamma + \kappa + \gamma^*}{2} v \end{cases}$$

This is $\dot{\mathbf{x}} = M\mathbf{x}$ with $\mathbf{x} = (\rho_{ee}, \rho_{cc}, v)^\top$ and

$$M = \begin{pmatrix} -\gamma & 0 & -2g \\ 0 & -\kappa & 2g \\ g & -g & -\frac{\gamma + \kappa + \gamma^*}{2} \end{pmatrix}$$

and initial condition $\mathbf{x}(0) = (1, 0, 0)^\top$. To get β , define $\mathbf{X} = \int_0^\infty \mathbf{x}(t) dt$. Integrating $\dot{\mathbf{x}} = M\mathbf{x}$ directly gives $M\mathbf{X} = -\mathbf{x}(0)$, since $\mathbf{x}(\infty) = 0$. M is invertible because $|g, 0\rangle$ is not tracked here, so no zero eigenvalue appears.

[expand on this](#)

Solving this system gives is simple numerically, we get $X_{cc} = \int_0^\infty \rho_{cc}(t) dt$ and

$$\beta = \kappa X_{cc} \tag{14}$$

As a sanity check, Fig. 2 shows the efficiency computed this way, and it is identical to the result obtained via the previous method.

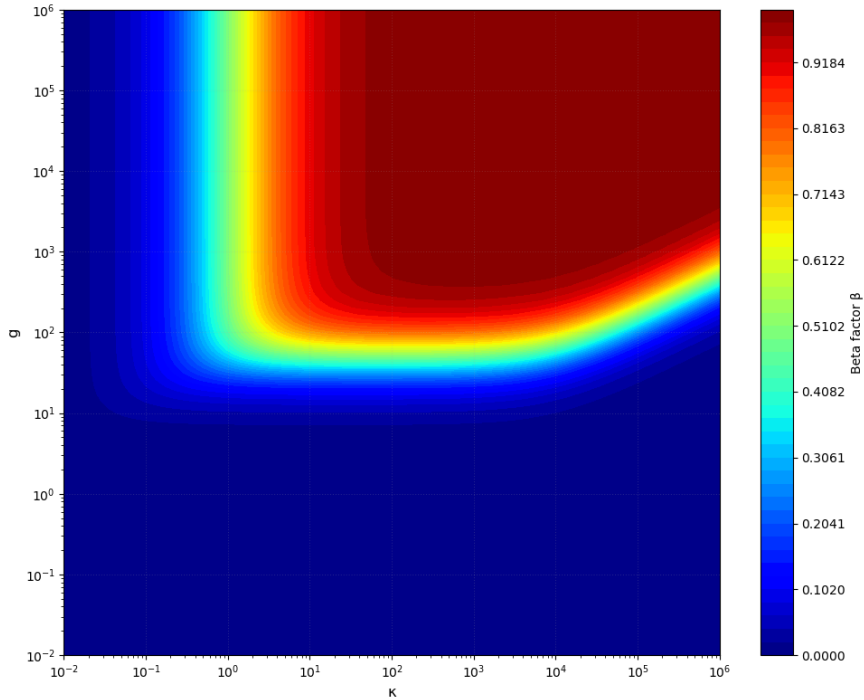


Figure 2: Efficiency β computed via the reduced 3×3 system.

4 Indistinguishability

Having computed the efficiency analytically, we now turn to the indistinguishability.

4.1 Hong-Ou-Mandel indistinguishability

For a single-photon emitter to be useful in quantum information applications, it is not enough that it emits one photon at a time: successive photons must also be identical, i.e. indistinguishable. Indistinguishability quantifies how well two photons from the same source can interfere, measured by the Hong-Ou-Mandel experiment.

4.1.1 The Hong-Ou-Mandel effect

The Hong-Ou-Mandel (HOM) experiment probes the quantum indistinguishability of two single photons by sending them onto a 50/50 beam splitter (BS) and monitoring coincidences at the two output ports.

The setup is shown in Figure 3: two photons enter the BS through input modes $\hat{a}_1$ and $\hat{a}_2$, and exit through output modes $\hat{a}_3$ and $\hat{a}_4$. For a symmetric 50/50 BS, the input and output modes are related by:

$$\hat{a}_3 = \frac{1}{\sqrt{2}}(\hat{a}_1 + \hat{a}_2), \quad \hat{a}_4 = \frac{1}{\sqrt{2}}(\hat{a}_1 - \hat{a}_2) \quad (15)$$

Inverting these relations gives $\hat{a}_1^\dagger = (\hat{a}_3^\dagger + \hat{a}_4^\dagger)/\sqrt{2}$ and $\hat{a}_2^\dagger = (\hat{a}_3^\dagger - \hat{a}_4^\dagger)/\sqrt{2}$. The two-photon input state $|1, 1\rangle \equiv \hat{a}_1^\dagger \hat{a}_2^\dagger |0, 0\rangle$ is therefore mapped to:

$$\begin{aligned} \hat{a}_1^\dagger \hat{a}_2^\dagger |0, 0\rangle &= \frac{1}{2}(\hat{a}_3^\dagger + \hat{a}_4^\dagger)(\hat{a}_3^\dagger - \hat{a}_4^\dagger) |0, 0\rangle \\ &= \frac{1}{2}(\hat{a}_3^{\dagger 2} - \hat{a}_3^\dagger \hat{a}_4^\dagger + \hat{a}_4^\dagger \hat{a}_3^\dagger - \hat{a}_4^{\dagger 2}) |0, 0\rangle \end{aligned} \quad (16)$$

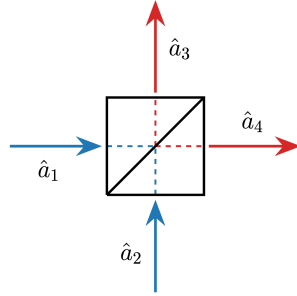


Figure 3: Schematic of the HOM experiment.

The four terms correspond to the four possible scattering outcomes, illustrated in Figure 4:

- (1) $\hat{a}_1$ reflected, $\hat{a}_2$ transmitted $\rightarrow$ both exit in mode $\hat{a}_3$
- (2) both photons transmitted $\rightarrow$ one photon in each output
- (3) both photons reflected $\rightarrow$ one photon in each output
- (4) $\hat{a}_1$ transmitted, $\hat{a}_2$ reflected $\rightarrow$ both exit in mode $\hat{a}_4$

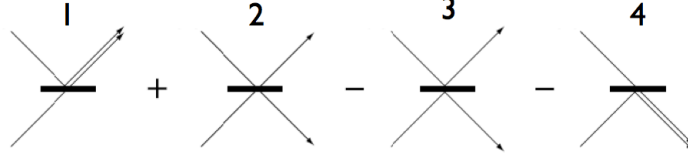


Figure 4: The four scattering amplitudes contributing to the two-photon output.

Crucially, for bosons the operators commute: $\hat{a}_3^\dagger \hat{a}_4^\dagger = \hat{a}_4^\dagger \hat{a}_3^\dagger$. Outcomes 2 and 3 therefore contribute amplitudes $+1/2$ and $-1/2$ to the final state $|1, 1\rangle$ and cancel completely. What remains is:

$$\hat{a}_1^\dagger \hat{a}_2^\dagger |0, 0\rangle = \frac{1}{2} (\hat{a}_3^{\dagger 2} - \hat{a}_4^{\dagger 2}) |0, 0\rangle = \frac{1}{\sqrt{2}} (|2, 0\rangle - |0, 2\rangle) \quad (17)$$

Both photons always exit the same port: the probability of a coincidence detection, one photon at $\hat{a}_3$ and one at $\hat{a}_4$, is exactly zero. This is the HOM dip.

4.1.2 Physical meaning

Let us briefly discuss the physical intuition behind the HOM effect. This cancellation is a manifestation of two-photon quantum interference. For identical photons there is no way to determine which photon was reflected and which was transmitted: outcomes 2 and 3 are indistinguishable as they both lead to the same final state $|1, 1\rangle$. Their probability amplitudes carry opposite signs and must be added coherently: they interfere destructively.

By contrast, outcomes 1 ($|2, 0\rangle$) and 4 ($|0, 2\rangle$) each arise from a single unambiguous path with no partner to interfere with. They survive and each occur with probability $1/2$: the photons always bunch into the same output port.

Note that this picture breaks down as soon as the photons become distinguishable. Any difference in an internal degree of freedom (frequency, temporal profile...) makes the path information accessible in principle, the coincidence paths are no longer identical, and the destructive interference is partial or absent. The coincidence probability then rises towards the classical value of $1/2$, and the visibility of the HOM dip becomes a direct quantitative measure of photon indistinguishability.

4.1.3 Simple simulation

To illustrate the HOM effect, we model each photon as a Gaussian wavepacket of width σ . The coincidence probability as a function of delay τ is:

$$P_{\text{coinc}}(\tau) = \frac{1}{2} \left(1 - |\langle \psi_1 | \psi_2(\tau) \rangle|^2 \right) \quad (18)$$

Figure 5 shows the plot from the simulation.

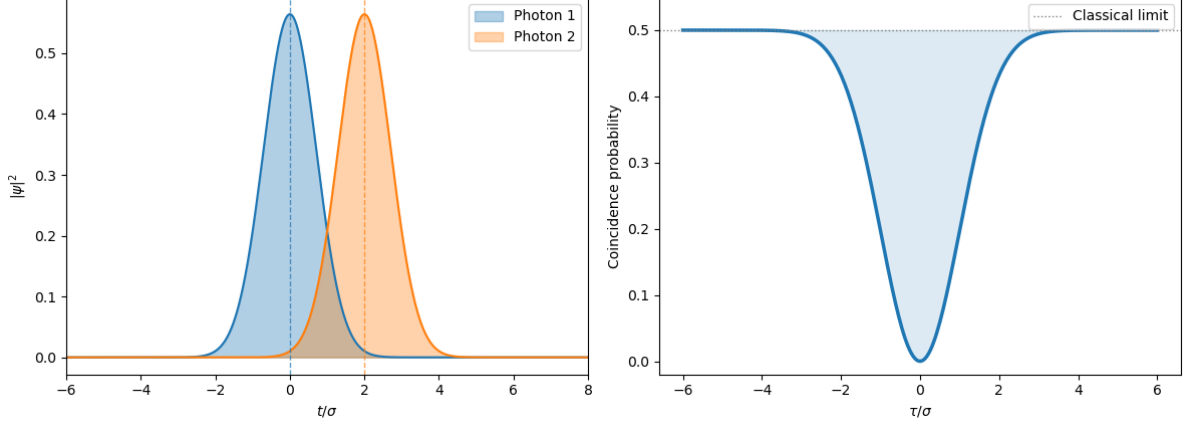


Figure 5: Left: two Gaussian input wavepackets at a relative delay $\tau = 2\sigma$. Right: coincidence probability for identical photons as a function of time delay.

Note that this is a very simplified simulation. We assume both photons have the same width σ and neglect several effects present in a real experiment, such as polarisation mismatch, chirp, and multi-photon contamination, all of which would partially fill the dip.

As an illustration of how photon parameters affect the coincidence probability, Figure 6 shows the effect of a frequency mismatch $\Delta\omega$. For Gaussian wavepackets, the overlap evaluates to:

$$|\langle\psi_1|\psi_2(\tau)\rangle|^2 = e^{-\tau^2/2\sigma^2} e^{-\Delta\omega^2\sigma^2/2} \quad (19)$$

so a non-zero $\Delta\omega$ raises the dip minimum from 0 to $\frac{1}{2}(1 - e^{-\Delta\omega^2\sigma^2/2})$.

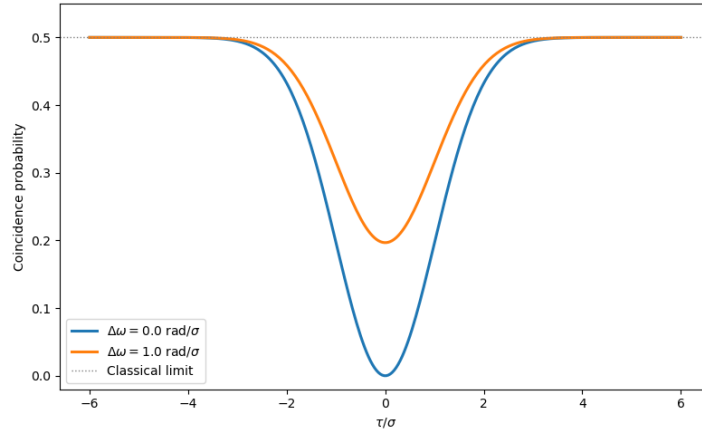


Figure 6: Coincidence probability for two values of frequency mismatch $\Delta\omega$.

As expected, we can see that a non-zero $\Delta\omega$ reduces the wavepacket overlap, partially restoring coincidences and raising the dip minimum above zero.

4.1.4 Deriving an expression for the indistinguishability

We now want to turn the HOM experiment into a quantitative measure. Define the normalised coincidence probability p_c as the probability of detecting one photon in each output port, normalised by the total photon flux in each port:

$$p_c = \frac{\int dt \int d\tau g_{34}^{(2)}(t, \tau)}{\left[\int dt \langle \hat{a}_3^\dagger(t) \hat{a}_3(t) \rangle \right] \left[\int dt \langle \hat{a}_4^\dagger(t) \hat{a}_4(t) \rangle \right]} \quad (20)$$

where $g_{34}^{(2)}(t, \tau) = \langle \hat{a}_3^\dagger(t) \hat{a}_4^\dagger(t + \tau) \hat{a}_3(t) \hat{a}_4(t + \tau) \rangle$ is the cross-correlation between the two output ports. The indistinguishability is then defined as $I = 1 - 2p_c$: it equals 1 when photons always bunch ($p_c = 0$) and 0 for classical light ($p_c = 1/2$).

We now derive $g_{34}^{(2)}$ explicitly. Substituting the beam splitter transformation (eq. 15) into $g_{34}^{(2)}$ and expanding the product of four operators yields 16 terms. Since we work with single-photon inputs, any term containing $\hat{a}_k^2$ or $(\hat{a}_k^\dagger)^2$ vanishes. Only four terms survive [2]:

$$\begin{aligned} g_{34}^{(2)}(t, \tau) = & \frac{1}{4} [\langle \hat{a}_1^\dagger(t) \hat{a}_2^\dagger(t + \tau) \hat{a}_2(t + \tau) \hat{a}_1(t) \rangle \\ & + \langle \hat{a}_2^\dagger(t) \hat{a}_1^\dagger(t + \tau) \hat{a}_1(t + \tau) \hat{a}_2(t) \rangle \\ & - \langle \hat{a}_1^\dagger(t) \hat{a}_2^\dagger(t + \tau) \hat{a}_1(t + \tau) \hat{a}_2(t) \rangle \\ & - \langle \hat{a}_2^\dagger(t) \hat{a}_1^\dagger(t + \tau) \hat{a}_2(t + \tau) \hat{a}_1(t) \rangle] \end{aligned} \quad (21)$$

We now apply two assumptions. **Identical statistics** means both photons are drawn from the same source and have the same single-photon correlation functions: $\langle \hat{a}_1^\dagger(t) \hat{a}_1(t') \rangle = \langle \hat{a}_2^\dagger(t) \hat{a}_2(t') \rangle \equiv \langle \hat{a}^\dagger(t) \hat{a}(t') \rangle$. In practice this means both photons have the same spectrum, linewidth, and temporal envelope — they are statistically identical copies.

No phase correlations means the two successive emissions are independent events: there is no fixed phase relationship between the field of photon 1 and the field of photon 2. Mathematically, any expectation value mixing operators from ports 1 and 2 factorises: $\langle \hat{a}_1^\dagger(\dots) \hat{a}_2(\dots) \rangle = \langle \hat{a}_1^\dagger(\dots) \rangle \langle \hat{a}_2(\dots) \rangle = 0$, since single-photon fields have zero mean. This is what allows the four-operator mixed expectation values to split into products of two-operator ones.

Under these two assumptions the first term becomes:

$$\langle \hat{a}_1^\dagger(t) \hat{a}_2^\dagger(t + \tau) \hat{a}_2(t + \tau) \hat{a}_1(t) \rangle = \langle \hat{a}^\dagger(t) \hat{a}(t) \rangle \langle \hat{a}^\dagger(t + \tau) \hat{a}(t + \tau) \rangle$$

and the third term is:

$$\langle \hat{a}_1^\dagger(t) \hat{a}_2^\dagger(t + \tau) \hat{a}_1(t + \tau) \hat{a}_2(t) \rangle = \langle \hat{a}_1^\dagger(t) \hat{a}_1(t + \tau) \rangle \langle \hat{a}_2^\dagger(t + \tau) \hat{a}_2(t) \rangle = |\langle \hat{a}^\dagger(t + \tau) \hat{a}(t) \rangle|^2$$

The two first terms are equal, and so are the two last. The factor of $\frac{1}{4}$ combines with the contributions and gives an overall $\frac{1}{2}$, and eq. (21) becomes:

$$g_{34}^{(2)}(t, \tau) = \frac{1}{2} [\langle \hat{a}^\dagger(t) \hat{a}(t) \rangle \langle \hat{a}^\dagger(t + \tau) \hat{a}(t + \tau) \rangle - |\langle \hat{a}^\dagger(t + \tau) \hat{a}(t) \rangle|^2] \quad (22)$$

Substituting into p_c and using $I = 1 - 2p_c$, the classical contributions (first term) cancel and one obtains:

$$I = \frac{\int_0^\infty dt \int_0^\infty d\tau |\langle \hat{a}^\dagger(t + \tau) \hat{a}(t) \rangle|^2}{\int_0^\infty dt \int_0^\infty d\tau \langle \hat{a}^\dagger(t) \hat{a}(t) \rangle \langle \hat{a}^\dagger(t + \tau) \hat{a}(t + \tau) \rangle} \quad (23)$$

4.2 The approach

Following the failure of the numerical approach to compute the efficiency, it is preferable to avoid computing I by naively evaluating the integrals numerically via quadrature. Even if it worked, this would be slow and subject to discretisation error.

Instead, this section is dedicated to solving it analytically, which is more involved than the efficiency but thankfully possible. Since the full derivation is quite long, we first give the conceptual approach to solving this.

The key insight is that the correlators in I can be written as sums of exponentials in t and τ , which makes the double integrals tractable in closed form. The strategy to get there has three steps:

1. **Density matrix.** We solve the Lindblad master equation analytically via eigendecomposition of the Liouvillian. This gives $\rho(t)$, useful for computing both the single-time correlators $\langle \hat{a}^\dagger(t)\hat{a}(t) \rangle$ and the two-time correlator $\langle \hat{a}^\dagger(t+\tau)\hat{a}(t) \rangle$.
2. **Green's function.** The two-time correlator $\langle \hat{a}^\dagger(t+\tau)\hat{a}(t) \rangle$ in the numerator requires knowing how the field propagates forward in time by τ . This is given by the retarded Green's function.
3. **Integration.** Once both quantities are written as sums of exponentials, the double integral over t and τ reduces to a finite sum of terms of the form $\int_0^\infty e^{\lambda t} dt = -1/\lambda$, giving an exact closed-form result for I .

4.3 Density matrix

Our starting point is once again the Lindblad master equation.

4.3.1 Vectorisation formalism

To solve it analytically, we once again use the vectorisation formalism, which converts the Lindblad equation into a linear ordinary differential equation.

Note however that we now work in the 2D basis $\{|e, 0\rangle, |g, 1\rangle\}$. Unlike the efficiency calculation, we do not need to track the state $|g, 0\rangle$ here. Physically, this is because once all population has decayed to the ground state with an empty cavity, it contributes nothing further to the cavity field correlations we measure.

In the Hilbert space, the density matrix is 2×2 :

$$\rho = \begin{pmatrix} \rho_{ee} & \rho_{ec} \\ \rho_{ce} & \rho_{cc} \end{pmatrix}$$

The vectorisation procedure flattens this into a column vector.

$$|\rho\rangle\rangle = \begin{pmatrix} \rho_{ee} \\ \rho_{cc} \\ \text{Re}(\rho_{ec}) \\ \text{Im}(\rho_{ec}) \end{pmatrix} \in \mathbb{R}^4$$

We note here that since $\rho_{ce} = \rho_{ec}^*$ (hermiticity), the two off-diagonal elements are not linearly independent. To keep our matrices real (which is numerically cleaner), we decompose the complex coherence into its real and imaginary parts.

The Lindblad equation in vectorised form becomes:

$$\frac{d}{dt} |\rho\rangle\rangle = \hat{L} |\rho\rangle\rangle$$

where $\hat{L}$ is the 4×4 Liouvillian superoperator matrix. The formal solution is:

$$|\rho(t)\rangle\rangle = e^{\hat{L}t} |\rho(0)\rangle\rangle$$

with the initial condition $|\rho(0)\rangle\rangle = (1, 0, 0, 0)^T$, corresponding to the pure state $|e, 0\rangle \langle e, 0|$.

4.3.2 Eigendecomposition

To evaluate the matrix exponential $e^{\hat{L}t}$ analytically, we use eigendecomposition. If $\hat{L} = V\Lambda V^{-1}$ where V contains the eigenvectors and Λ is the diagonal matrix of eigenvalues, then:

$$e^{\hat{L}t} = V e^{\Lambda t} V^{-1} \tag{24}$$

The diagonal exponential is trivial:

$$e^{\Lambda t} = \begin{pmatrix} e^{\lambda_1 t} & 0 & 0 & 0 \\ 0 & e^{\lambda_2 t} & 0 & 0 \\ 0 & 0 & e^{\lambda_3 t} & 0 \\ 0 & 0 & 0 & e^{\lambda_4 t} \end{pmatrix}$$

The product $V e^{\Lambda t} V^{-1}$ can be written explicitly as a spectral decomposition. Since the diagonal exponential decomposes $e^{\Lambda t} = \sum_i e^{\lambda_i t} |e_i\rangle\langle e_i|$ in the eigenbasis, we obtain:

$$\begin{aligned} e^{\hat{L}t} &= V e^{\Lambda t} V^{-1} \\ &= \sum_i e^{\lambda_i t} (V |e_i\rangle)(\langle e_i| V^{-1}) \\ &= \sum_i e^{\lambda_i t} |v_i\rangle\langle w_i| \end{aligned} \tag{25}$$

and so it follows that

$$|\rho(t)\rangle\rangle = \sum_{i=1}^4 e^{\lambda_i t} |v_i\rangle\langle w_i|\rho(0)\rangle\rangle = \sum_{i=1}^4 e^{\lambda_i t} \xi_i |v_i\rangle \tag{26}$$

where $|v_i\rangle$ are the right eigenvectors of $\hat{L}$ (columns of V), $\langle w_i|$ the left eigenvectors of $\hat{L}$ (rows of V^{-1}) and ξ_i the overlap. The density matrix thus evolves as a superposition of four independent decay channels, each with rate λ_i . From the evolved vectorised density matrix, we extract the physical density matrix elements:

$$\rho_{ee}(t), \rho_{cc}(t), \rho_{ec}(t) = \text{Re}(\rho_{ec}(t)) + i \cdot \text{Im}(\rho_{ec}(t)) \tag{27}$$

These density matrix elements are essential for the next step. The cavity population $\rho_{cc}(t)$ directly gives the single-time intensity correlator $\langle \hat{a}^\dagger(t) \hat{a}(t) \rangle = \rho_{cc}(t)$, which is needed for the denominator of the indistinguishability integral. More importantly, both $\rho_{cc}(t)$ and $\rho_{ec}(t)$ will be combined with the Green's function to compute the two-time correlators $\langle \hat{a}^\dagger(t + \tau) \hat{a}(t) \rangle$.

4.4 Green's Function

To compute the two-time correlators, we need to understand how the cavity field evolves across a time delay. This is where the retarded Green's function comes in.

Conceptually, $\hat{G}^R(\tau)$ is a single-particle propagator: it describes how a quantum amplitude evolves forward in time by τ . Combined with the density matrix at time t via the Kadanoff-Baym equation, it will give us access to the two-time field correlations $\langle \hat{a}^\dagger(t + \tau) \hat{a}(t) \rangle$.

4.4.1 Green function definition

In the frequency domain, the retarded Green's function satisfies the Dyson equation [3]:

$$(\omega - H - \Sigma^R) \hat{G}^R(\omega) = I \tag{28}$$

where H is the Hamiltonian and Σ^R encodes the dissipative self-energies (decay rates). Inverting gives [2]:

$$\hat{G}^R(\omega) = \begin{pmatrix} \omega + i\gamma/2 + i\gamma^*/2 & g \\ g & \omega - \delta + i\kappa/2 \end{pmatrix}^{-1} \tag{29}$$

However, we are interested in Green's function in the time domain. It satisfies the equation of motion [2] (for $\tau > 0$):

$$\begin{aligned} i \frac{\partial \hat{G}^R(\tau)}{\partial \tau} &= i\delta(\tau)\hat{I} + [H + \Sigma^R(0)]\hat{G}^R(\tau) \\ &= [H + \Sigma^R(0)]\hat{G}^R(\tau) \end{aligned} \quad (30)$$

Define the propagation matrix:

$$M = -i[H + \Sigma^R(0)] = \begin{pmatrix} -(\gamma + \gamma^*)/2 & -ig \\ -ig & -i\delta - \kappa/2 \end{pmatrix} \quad (31)$$

Then the equation of motion becomes $\partial \hat{G}^R(\tau)/\partial \tau = M\hat{G}^R(\tau)$, which is solved by:

$$\hat{G}^R(\tau) = e^{M\tau} \quad (32)$$

with the boundary condition $\hat{G}^R(0^+) = I$ because the field at time t is fully correlated with itself.

4.4.2 Eigendecomposition

This is exactly the same form as the density matrix evolution, we can evaluate $e^{M\tau}$ analytically using eigendecomposition. Diagonalizing the 2×2 propagation matrix $M = U\tilde{\Lambda}U^{-1}$, where $\tilde{\Lambda}$ contains the eigenvalues μ_i :

$$\hat{G}^R(\tau) = \begin{pmatrix} G_{ee}^R(\tau) & G_{ec}^R(\tau) \\ G_{ce}^R(\tau) & G_{cc}^R(\tau) \end{pmatrix} = e^{M\tau} = Ue^{\tilde{\Lambda}\tau}U^{-1} = \sum_{j=1}^2 e^{\mu_j\tau} |v_j\rangle \langle w_j| \quad (33)$$

where $|v_j\rangle$ are the right eigenvectors and $\langle w_j|$ are the left eigenvectors of M .

Similarly to the density matrix, we managed to express Green's function as a sum of exponentials.

4.5 Two-time correlator

Armed with these results, we can now finally compute the two-time correlator.

4.5.1 Kadanoff-Baym relation

The connection between the density matrix at time t , the retarded Green function and the field correlation is given by the Kadanoff-Baym equations [3] from non-equilibrium Green's function theory.

This relation reads:

$$\hat{G}^<(t, t + \tau) = \hat{G}^R(\tau)\hat{\rho}(t) \quad (34)$$

where $\hat{G}^<$ is the lesser Green's function encoding the two-time correlations [2]:

$$\hat{G}^<(t, t + \tau) = \begin{pmatrix} \langle \hat{\sigma}^\dagger(t + \tau)\hat{\sigma}(t) \rangle & \langle \hat{\sigma}^\dagger(t + \tau)\hat{a}(t) \rangle \\ \langle \hat{a}^\dagger(t + \tau)\hat{\sigma}(t) \rangle & \langle \hat{a}^\dagger(t + \tau)\hat{a}(t) \rangle \end{pmatrix} \quad (35)$$

From this, the two-time correlator of the cavity field can be extracted by taking the cavity-cavity component of the matrix:

$$\begin{aligned} \langle \hat{a}^\dagger(t + \tau)\hat{a}(t) \rangle &= \langle g, 1 | \hat{G}^R(\tau)\hat{\rho}(t) | g, 1 \rangle \\ &= G_{cc}^R(\tau)\rho_{cc}(t) + G_{ce}^R(\tau)\rho_{ec}(t) \end{aligned} \quad (36)$$

4.5.2 Notes about the components

Notice that having the vector $|\rho(t)\rangle\rangle$ or the matrix $G^R(\tau)$ as sum of exponential means that we are also able to express each individual elements as a sum.

For the sake of completeness, let us express the quantities of interest given in eq. (36).

To start with $|\rho(t)\rangle\rangle$, we go back to equation (26) and can write:

$$\begin{aligned}\rho_{cc}(t) &= \sum_i \xi_i [v_i]_2 e^{\lambda_i t} \\ \rho_{ec}(t) &= \sum_i \xi_i ([v_i]_3 + i[v_i]_4) e^{\lambda_i t}\end{aligned}\tag{37}$$

where $[v_i]_k$ denotes the k -th component of $|v_i\rangle\rangle$. Similarly, for $\hat{G}^R$: since $\hat{G}^R(\tau) = \sum_j e^{\mu_j \tau} |v_j\rangle \langle w_j|$, each matrix element reads:

$$\begin{aligned}G_{cc}^R(\tau) &= \sum_j [v_j]_2 [w_j]_2 e^{\mu_j \tau} \\ G_{ce}^R(\tau) &= \sum_j [v_j]_2 [w_j]_1 e^{\mu_j \tau}\end{aligned}\tag{38}$$

where $[v_j]_k$ and $[w_j]_k$ denote the k -th components of the right and left eigenvectors of M .

The sum in equation (36) can therefore be rewritten:

$$\boxed{\langle \hat{a}^\dagger(t + \tau) \hat{a}(t) \rangle = \sum_{i,j} \alpha_{ij} e^{\lambda_i t + \mu_j \tau}}\tag{39}$$

where α_{ij} collects the scalar prefactors from individual eigendecompositions. Note that these can be complex valued here.

4.6 Integration

Finally, we can use our correlators written in this new form (as sums of exponentials) to evaluate their integrals by exploiting the closed form:

$$\int_0^\infty e^{\lambda t} dt = -\frac{1}{\lambda}\tag{40}$$

4.6.1 Validity of the closed form

Note that equation (40) is valid under the condition that $\text{Re}(\lambda) < 0$.

We know the Liouvillian describe a dissipative system, where the eigenvalues describe the evolution of eigenmodes, that need to decay to not violate energy conservation.

Similarly, the retarded Green's function describes the propagation of the field's correlation, that decay to 0 when τ goes to infinity. The eigenvalues therefore have negative real parts. We can also notice that the propagation matrix M is constructed with dissipative terms on the diagonal. We can therefore apply the closed form without worry.

4.6.2 Numerator

The numerator of the indistinguishability is the two-time correlator and follows from 39:

$$\begin{aligned}\int_0^\infty dt \int_0^\infty d\tau |\langle \hat{a}^\dagger(t + \tau) \hat{a}(t) \rangle|^2 &= \int_0^\infty dt \int_0^\infty d\tau \left| \sum_{i,j} \alpha_{ij} e^{\lambda_i t + \mu_j \tau} \right|^2 \\ &= \sum_{i,j,k,l} \alpha_{ij}^* \alpha_{kl} \int_0^\infty e^{(\lambda_i + \lambda_k^*)t} dt \int_0^\infty e^{(\mu_j + \mu_l^*)\tau} d\tau \\ &= \sum_{i,j,k,l} \frac{\alpha_{ij}^* \alpha_{kl}}{(\lambda_i + \lambda_k^*)(\mu_j + \mu_l^*)}\end{aligned}\tag{41}$$

4.6.3 Denominator

The denominator is simpler because it involves only single-time correlators. The two factors $\langle \hat{a}^\dagger(t)\hat{a}(t) \rangle$ and $\langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau) \rangle$ are both just the instantaneous cavity population at a given time, which is directly ρ_{cc} evaluated at that time:

$$\langle \hat{a}^\dagger(t)\hat{a}(t) \rangle = \rho_{cc}(t) = \sum_i \xi_i [v_i]_2 e^{\lambda_i t} \quad (42)$$

The denominator integral therefore factorises into a product of two independent integrals:

$$\begin{aligned} \int_0^\infty dt \int_0^\infty d\tau \langle \hat{a}^\dagger(t)\hat{a}(t) \rangle \langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau) \rangle &= \int_0^\infty \rho_{cc}(t) dt \int_0^\infty \rho_{cc}(\tau) d\tau \\ &= \left(\int_0^\infty \sum_i \xi_i [v_i]_2 e^{\lambda_i t} dt \right)^2 \\ &= \left(\sum_i \frac{\xi_i [v_i]_2}{\lambda_i} \right)^2 \end{aligned} \quad (43)$$

4.7 Plot

This derivation finally allows us to bypass the time integration. Figure 7 shows the indistinguishability computed this way.

Note that from a computational aspect, this method is both exact and efficient across the parameter space (the simulation time is constant regardless of the values of κ or g).

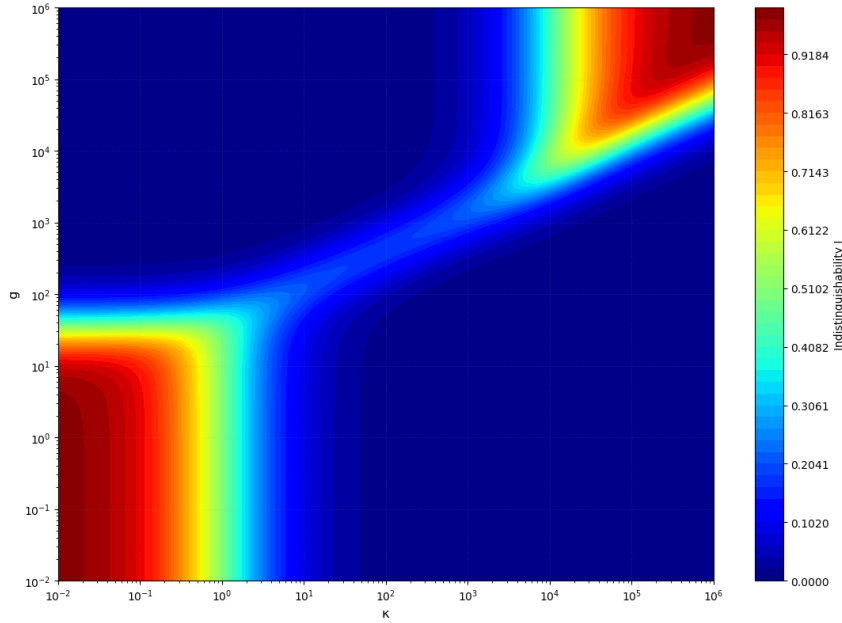


Figure 7: Indistinguishability I as a function of the coupling g and cavity decay rate κ , for $\gamma^*/\gamma = 10^4$.

4.8 Discussion on the different regimes

Let us now take a step back, and remember our initial goal. The reason we were looking for the efficiency and indistinguishability of these emitter-cavity systems was to find a new regime that isn't currently exploited in single photon emitters designs. We now have a tool to compute I and β for any set of parameters γ, γ^* .

talk about the different regimes in the Grange paper if it is useful

5 Litterature examples

CURRENT STATE OF THE ART - where are the current state of the art cavities? - place them in the I,beta plots and discuss - what values of kappa are archievable? we want moderate g but low kappa - explain that I,beta results are very general - apply to all types of emitters - some discussion on how to get the values from the wavelength for example in Annex - explain the 3 types of cavity that we have

#	System
1	GaN QDs in a deep-UV resonant cavity nanopillar (AlGaN/AlN DBR) [4]
3	GaN photonic crystal nanobeam laser [5]
4	GaN L3 photonic crystal cavity on silicon [6]
5	GaN photonic crystal cavity on silicon (nonlinear conversion) [7]
6	GaN microring resonator (photon pair source) [8]

Table 1: Emitter-cavity systems from the literature.

#	g (GHz)	κ (GHz)	γ (GHz)	γ^* (GHz)	β	I
1		744	0.5	2838		
3		351	0.5	2838		
4		10.2	5	1637		
5		4.5	5	1637		
6		0.45	5	1637		

Table 2: Parameters and computed figures of merit for the systems in Table 1.

The detail of how the quantities in table 2 are computed is given in Appendix C.

PUCCELL EFFECT POSSIBLE? MAYBE NOT DUE TO HIGH GAMMA* - would large pure dephasing suppress the purcell effect?

6 Approximations

7 Geometries

8 Conclusion

A Analytical derivation of the efficiency

B Relation between parameters of the QE-cavity system

The parameters used in our simulations are not always directly reported in the literature. It is therefore useful to know how to convert between measured quantities and the ones relevant here.

B.1 Cavity decay rate and quality factor

$$Q = \frac{\omega_0}{\kappa} \quad (44)$$

where:

- ω_0 is the resonance frequency of the cavity,
- κ is the cavity decay rate (linewidth, in angular frequency units),
- Q is the quality factor.

B.2 Linewidth and dephasing

The spectral linewidth of the emitter is related to the coherence time T_2 by:

$$\text{FWHM} = \frac{1}{T_2} = \frac{c \Delta\lambda}{\lambda^2} \quad (45)$$

where $\Delta\lambda$ is the linewidth measured in wavelength units and λ is the central wavelength. The total dephasing rate $1/T_2$ has contributions from both radiative decay and pure dephasing:

$$\text{FWHM} = \frac{1}{T_1} + \frac{2}{T_2^*} = \gamma + \gamma^* \quad (46)$$

where:

- T_2 is the total coherence time,
- T_1 is the excited state lifetime (radiative decay), with $\gamma = 1/T_1$,
- T_2^* is the pure dephasing time, with $\gamma^* = 2/T_2^*$,

B.3 Unit conventions for coupling and decay rates

The parameters g , κ , γ , and γ^* appear in the literature under several different conventions, which can cause confusion when comparing values across papers.

- **Energy units** ($\hbar g$, $\hbar \kappa$ in eV or μeV).
- **Angular frequency** (g , κ in rad/s).
- **Ordinary frequency** ($g/2\pi$, $\kappa/2\pi$ in Hz). Often simply written as g or κ .

In this report, all rates are ordinary frequencies (i.e. divided by 2π , in Hz). For practical reference, $1 \mu\text{eV} \approx 241 \text{ MHz} \approx 0.241 \text{ GHz}$.

C Computing parameters from literature

The cavity decay rate follows directly from the quality factor and resonant wavelength, that are commonly given in the literature:

$$\kappa = \frac{c}{\lambda_0 Q} \quad (47)$$

#	Q	λ_0 (nm)	κ (GHz)
1	1 600	252	744
3	1 900	450	351
4	22 500	1305	10.2
5	44 000	1512	4.5
6	430 000	1550	0.45

Table 3: Cavity parameters extracted from the literature, used to compute κ .

For the values of γ and γ^* , we use typical values of emitters characterised at LASPE.

- Emitter A $\lambda = 650$ nm, $T_1 = 2$ ns, FWHM = 4 nm
- Emitter B $\lambda = 1350$ nm, $T_1 = 0.2$ ns, FWHM = 10 nm

Using equations (46) and (45), we find:

- Emitter A: $\gamma = 0.5$ GHz, $\gamma^* = 2838$ GHz.
- Emitter B: $\gamma = 5$ GHz, $\gamma^* = 1637$ GHz.

The choice of emitter A or B is simply decided by the cavity resonance frequency.

The coupling strength g is not directly reported in the literature for GaN systems. We therefore use typical values estimated from comparable cavity geometries, even when these are reported for other material systems. One can expect these values to not be exact, but it is the best we do to try to estimate the possibility of exploiting this unconventional regime.

Note that the analytical derivation we spent the first part doing also assumes perfect resonance match between the cavity and emitter, and we can see it's already not the case for us here, so this will be another approximation.

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